Color Classification on `pregnancy_MiniLM_L6`: Attention vs. Pooling, Data Features, and Relational GNNs

Experiment 08 summary

July 15, 2026

Setup

Nodes are pregnancy visits (61,046), colors are patients (6,590), and the graph is a kNN graph (k=20) over 64-dimensional MiniLM visit embeddings. Both tasks are binary: predicting the label per node and per color. Test patients are those of the dataset's original test file and are never used for model or hyperparameter selection; the remaining patients are split 80/20 into train/validation (seed 42), and each node inherits its patient's assignment. Every variant was tuned with its own Optuna study (selection on validation AUC only; $\sim$60–140 configurations per variant). All variants share the same node-level GNN task, loss, early stopping, and evaluation.

Results

Variant	What was done	Val AUC (node/color)	Test AUC (node/color)
rgat	Two-relation graph (kNN edges + same-patient edges), relational attention, MiniLM features as input	.884 / .868	**.819 / .800**
rgcn	Same two-relation graph with relational graph convolution instead of attention	.881 / .867	.785 / .785
feat	Standard GNN on the kNN graph with MiniLM features as input (no color embedding)	.684 / .859	.630 / .770
feat_color	Same as feat, with a learned color embedding concatenated to the features	.678 / .846	.619 / .770
pool	GNN on learned color embeddings; color prediction by mean-pooling node embeddings per color	.652 / .829	.614 / .773
attention	GNN on learned color embeddings; color prediction by a self-attention classifier over the color-embedding matrix	.639 / .575	.625 / .500

Table 1: Best configuration per variant, selected by validation AUC. Test AUC is reported on held-out patients.

Analysis

1. The attention classifier should be dropped. The attention head's color-task test AUC is 0.500 — exactly chance level (validation: 0.575). Replacing only the color head with mean-pooling of GNN node

embeddings per color, while keeping the same inputs and node task (*pool*), raises the color task to 0.773 test / 0.829 validation. The failure is structural: the embeddings of validation and test colors receive no gradient from the color loss, so a classifier reading only those embeddings has nothing generalizable to attend to. The pooled head avoids this because it reads the GNN output, which is shaped by the graph and the node-level loss for all nodes.

2. Direction 1 — data features help; the color embedding does not. Feeding the MiniLM visit embeddings to the GNN (*feat*) outperforms learned color embeddings (*pool*) on both tasks. However, concatenating the learned color embedding on top of the features (*feat_color*) is marginally *worse* than the features alone. The learned per-color embedding is dead weight: all the usable signal already resides in the data features and the graph structure.

3. Direction 2 — the same-color relation is the main win. Adding a second edge type connecting visits of the same patient, and processing the two relations with a relational GNN (*rgcn*/*rgat*), lifts node-level test AUC from ~ 0.63 to 0.79–0.82 and the color task to ~ 0.80 — about $+15$ AUC points over every non-relational variant. The result is robust: the top-5 configurations of each relational variant lie within ~ 0.005 validation AUC of each other. RGAT leads RGCN on test despite exploring fewer configurations, suggesting further headroom.

Recommendation

Adopt the two-relation formulation (RGAT, MiniLM features, no color embedding) as the main model; drop the attention classifier and the learned color embeddings. Next: extend the RGAT study (more trials) and validate the same comparison on the eight disease datasets.